// SENSORS WHO LIE

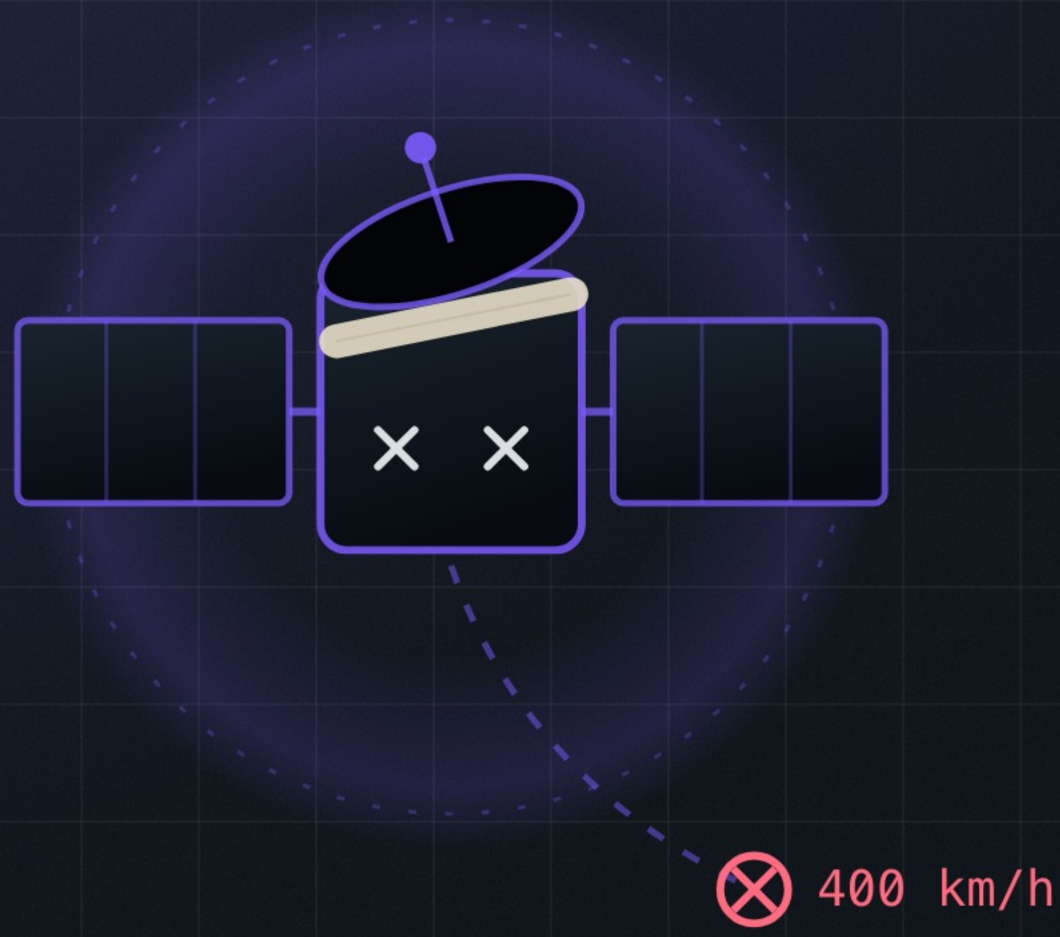
It worked perfectly. Then: traffic.

Our jitter filter was flawless right up until someone sat in Bangalore traffic.

It could not tell a parked car from one moving at walking pace.

// THE CAST

One number, every context



THE CONCUSSED WITNESS

exhibit 01

LocationManager

A single global threshold is the most confident wrong decision in your codebase. It swears the same thing is true while walking, cycling, driving, and standing still.

// THE CLAIM YOU MADE

A constant is an assumption

```
private fun minDisplacementForSpeed(speedMps: Double): Double = when {
    speedMps < 2.5 -> 2.0    // walking
    speedMps < 7.0 -> 3.0    // cycling
    else           -> 5.0    // driving
}
```

// THE FIX

Thresholds take context

01 **Function of speed band**
walking and motorway are not the same filter

02 **Function of time since last fix**
a 30 second gap changes what is plausible

03 **Function of recent history**
what was this device doing ten seconds ago

04 **Name the assumption in code**
if it is "all contexts", that is the bug

A constant with no context is a bug on a delay

When you write a number into a filter, ask which context you are assuming.

If the answer is "all of them", you have not written a threshold.